

End-of-module reflective activity

Time Series Analysis – guided self-assessment

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Instructions

This *ungraded* activity is intended to help you:

- identify which notions of the module you have mastered and which ones still need work;
- formalise what you retain from the six chapters;
- plan your revision for the final exam and the integrative project.

Take about **45 minutes** to answer in writing (handwritten or in LaTeX). Be honest: this document will not be collected and the benefit lies in the metacognitive effort you invest.

Question – 1. Self-positioning by chapter

For each chapter, give yourself a *perceived mastery* score from 1 (very fuzzy) to 5 (I could explain it to a classmate) and justify in one sentence.

Chapter	Score 1–5	Justification (one sentence)
Ch.1 – Basic statistics	_____	
Ch.2 – Decomposition	_____	
Ch.3 – Stationarity (ACF/PACF)	_____	
Ch.4 – Non-stationarity (ADF/KPSS)	_____	
Ch.5 – ARMA models	_____	
Ch.6 – ARCH/GARCH	_____	

Question – 2. The hardest notion and why

Identify the notion of the module that you found most *difficult*. Describe:

- what makes it difficult for you (algebra, intuition, probability, the link with the code...);
- one concrete strategy to work on it next week (revisit a specific example, redo a lab exercise, draw a diagram, re-derive a formula by hand...).

Question – 3. Sketch the Box–Jenkins workflow from memory

Without looking at the lecture notes, sketch the Box–Jenkins workflow from a raw series X_t all the way to a validated ARIMA forecast. Make sure your diagram includes:

raw X_t $\rightarrow$ stationarise $\rightarrow$ identify (p, q) $\rightarrow$ estimate $\rightarrow$ validate $\rightarrow$ forecast.

For each arrow, write the *tool* you would use (e.g. ADF, ACF, MLE, Ljung–Box, MA(∞) variance). Then compare with the *Box–Jenkins identification* section of Chapter 5 in the lecture

notes and correct any missing or misplaced step.

Question – 4. Quick mental computations

- (a) For $X_t = 0.7 X_{t-1} + \varepsilon_t$ with $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, write the values of $\rho(1), \rho(2), \rho(3)$. Is the PACF of this AR(1) larger than the ACF at lag 2? Why?
- (b) For an MA(1) process $X_t = \varepsilon_t + \theta \varepsilon_{t-1}$ with $\theta = 0.5$, what is $\rho(1)$? At which value of θ does the maximum of $|\rho(1)|$ occur?
- (c) For a GARCH(1,1) with $(\alpha_0, \alpha_1, \beta_1) = (0.05, 0.10, 0.85)$, compute the unconditional variance σ^2 and the half-life of a volatility shock (number of days τ such that $(\alpha_1 + \beta_1)^\tau = 1/2$).
- (d) You run an ADF test and obtain $\text{stat} = -1.5$, while KPSS gives $\text{stat} = 1.8$. Both at 5% levels. In which cell of the ADF/KPSS combination matrix does this land? What action would you take?

Question – 5. From maths to code

Pick *one* mathematical formula from the module and explain:

- how you *implemented* it in a lab (in NumPy / pandas / statsmodels / arch);
- one pitfall or bug you encountered (shape mismatch, off-by-one indexing, time-zone, log-difference vs simple difference, σ^2 versus σ , units (% vs. fraction)...);
- how you fixed it.

This question exercises the *maths* $\leftrightarrow$ *code* translation skill, which is one of the key competences of the module.

Question – 6. The thread that ties the six chapters

In 5–10 lines, explain why these six chapters are gathered into a single module. What is the *common thread*? What would be lost by studying any one of them in isolation?

Hint. Think of the role of the residual $\hat{\varepsilon}_t$, which:

1. appears in Chapter 2 as the residual of decomposition;
2. is the object whose stationarity is tested in Chapter 3 (ACF, Ljung–Box);
3. is the object the unit-root tests are run on in Chapter 4;
4. is what an ARMA model in Chapter 5 tries to make white;
5. is the object whose conditional variance is modelled in Chapter 6.

Question – 7. Connecting to the integrative project

Look back at the seven parts (A–G) of the final integrative project (`projet-final/sujet.pdf`). For each part, write the chapter it builds on:

Project part	Chapter mobilised
A – Descriptive statistics + JB test	
B – Decomposition + linear trend	
C – ACF/PACF + Ljung–Box	
D – ADF/KPSS	
E – ARMA selection	
F – GARCH(1, 1) + VaR	
G – Kupiec back-test (synthesis)	

What does the back-test in Part G say about *what we have not modelled*? In other words: even a passing Kupiec test is not a proof of model correctness – can you identify two limits of the joint ARMA + GARCH model that the back-test does not see?

Question – 8. Personal action plan

List *three* concrete actions you will take *this week* to consolidate your learning of the module:

1.
2.
3.

A concrete action has an action verb (*revisit, code, redo, sketch...*), a precise object, and a *timing* (when will you do it?).

Good thinking. This activity will not be collected: it is for you.

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